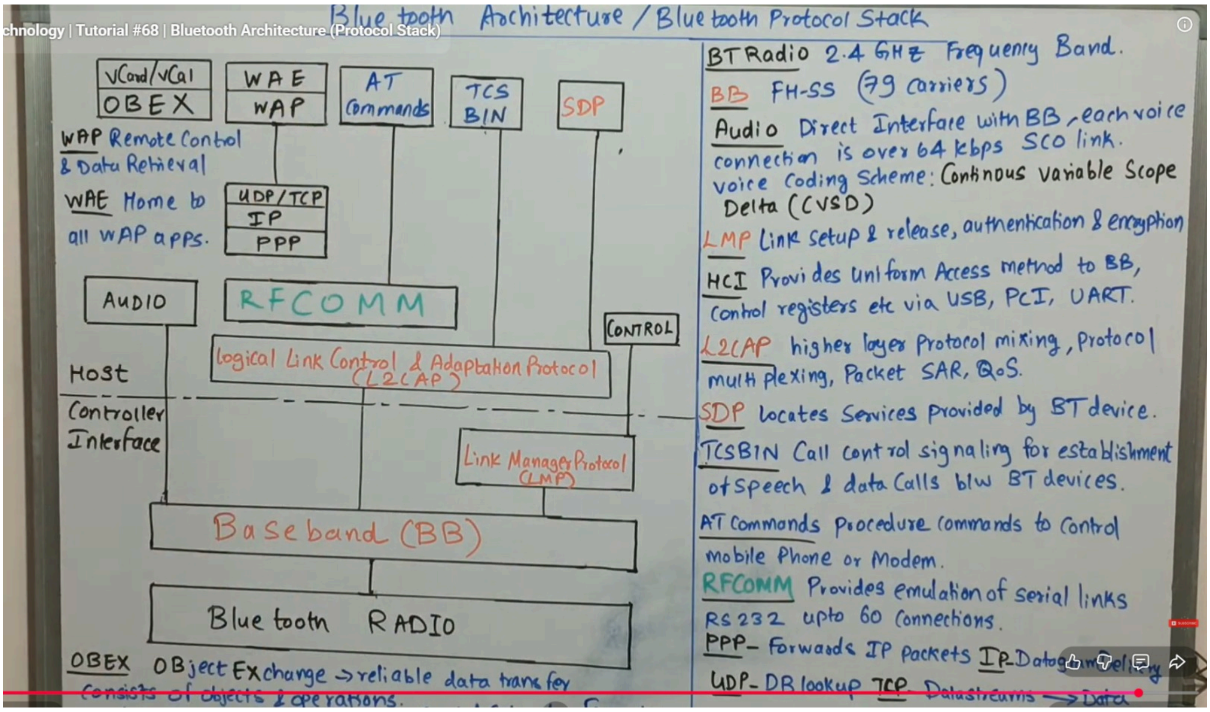

📡 Bluetooth Architecture / Protocol Stack

Introduction

Bluetooth architecture is divided into **layers** that manage communication between devices. It consists of **Controller part** and **Host part**, connected through **HCI (Host Controller Interface)**.

🔸 1. Bluetooth Radio Layer

- Operates in **2.4 GHz ISM band**
- Uses **Frequency Hopping Spread Spectrum (FHSS)**
- 79 channels used
- Responsible for **wireless transmission**

🔸 2. Baseband (BB)

- Handles **physical channel and link control**
- Supports:
  - **SCO (voice link)**
  - **ACL (data link)**
- Performs **error correction and packet formation**

🔸 3. Link Manager Protocol (LMP)

- Responsible for:
  - Link setup and release
  - Authentication
  - Encryption
  - Power control

🔸 4. Host Controller Interface (HCI)

- Interface between **host and controller**
- Provides uniform access via:
  - USB
  - UART
  - PCI

🔸 5. L2CAP (Logical Link Control and Adaptation Protocol)

- Multiplexes higher layer protocols
- Provides:
  - Segmentation and reassembly (SAR)
  - Quality of Service (QoS)

🔸 6. RFCOMM

- Emulates **serial port communication (RS-232)**
- Supports multiple connections

🔸 7. SDP (Service Discovery Protocol)

- Used to **discover services** offered by Bluetooth devices

🔸 8. TCS (Telephony Control Protocol)

- Used for **call control signalling**
- Establishes voice/data calls

🔸 9. AT Commands

- Used to control:
  - Mobile phone
  - Modem operations

🔸 10. OBEX (Object Exchange Protocol)

- Used for **file transfer**
- Ensures reliable data exchange

🔸 11. PPP, TCP/IP, UDP

- Enable **internet communication over Bluetooth**
- PPP → forwards IP packets
- TCP/UDP → transport protocols

🔸 12. Audio Layer

- Directly interfaces with baseband
- Used for **voice transmission**

🧠 Quick Summary (Exam Tip)

- **Radio + Baseband → Physical communication**
- **LMP → Link control & security**
- **HCI → Interface**
- **L2CAP → Data handling**
- **RFCOMM / SDP / OBEX → Applications**

📡 Frequency Reuse with Clustering

🔹 Definition

Frequency reuse is a technique in cellular networks where the **same frequency channels are reused in different cells** to improve spectrum efficiency and increase system capacity.

🔸 Basic Concept

- The total coverage area is divided into **small cells**
- Each cell is assigned a set of **frequencies**
- The same frequencies are reused in **non-adjacent cells** to avoid interference

---

🔸 Clustering Concept

- A group of adjacent cells using **different frequency sets** is called a **cluster**
- Each cluster uses the **complete set of available frequencies**
- Clusters are **repeated over the entire network**

🔸 Cluster Size (N)

Cluster size determines how frequencies are distributed.

N = i^2 + ij + j^2

Where:

- **i, j** = integers
- **N** = number of cells in a cluster

📌 Examples:

- (i=1, j=1) → N = 3
- (i=2, j=1) → N = 7
- (i=2, j=2) → N = 12

🔸 Frequency Reuse Pattern

- Cells using the same frequency are called **co-channel cells**
- They are separated by a distance to reduce interference

🔸 Reuse Distance (D)

D = \sqrt{3N} \cdot R

Where:

- **D** = distance between co-channel cells
- **R** = cell radius
- **N** = cluster size

🔸 Advantages

- Efficient use of limited frequency spectrum
- Increased network capacity
- Supports more users

🔸 Disadvantages

- Co-channel interference
- Complex frequency planning required

🧠 Quick Summary (Exam Tip)

- **Cluster = group of cells with unique frequencies**
- **Frequency reuse = reuse same frequencies in far cells**
- Larger **N** → **less interference, lower capacity**
- Smaller **N** → **more capacity, more interference**

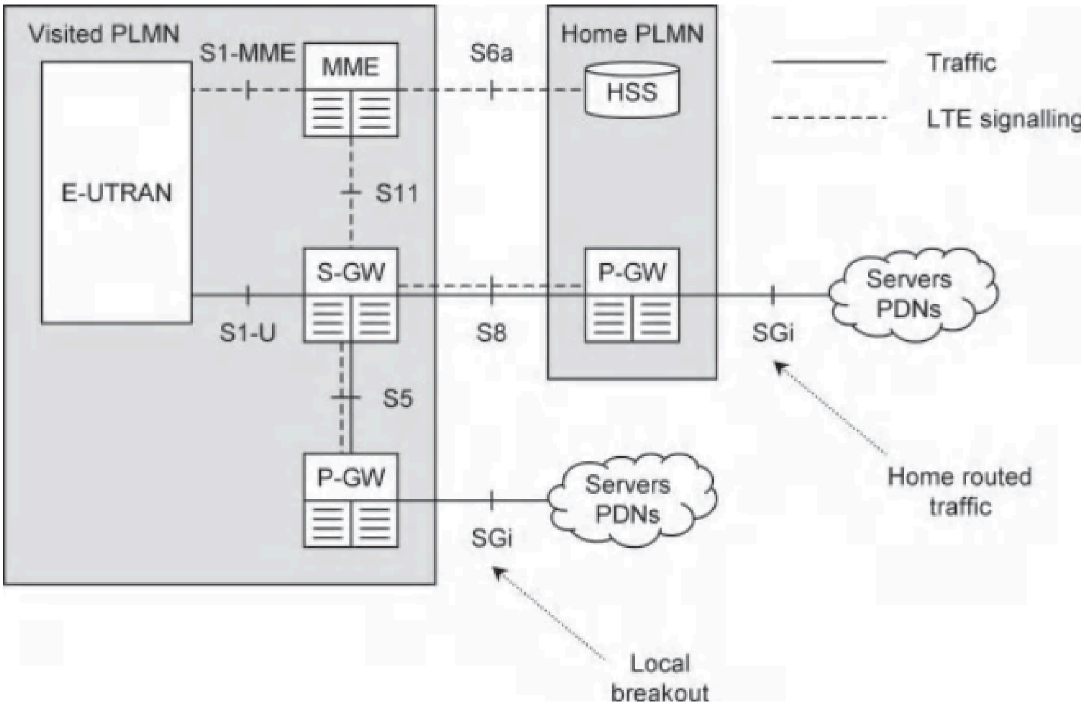

Figure 2.5 Architecture of LTE for a roaming mobile.

Here's a **clear exam-ready answer with diagram + explanation** of LTE architecture components:

📡 LTE Architecture Components

🔹 1. E-UTRAN (Evolved Universal Terrestrial Radio Access Network)

📌 Component: eNodeB (eNB)

- Main base station in LTE
- Connects **user equipment (UE)** to the network
- Handles:
  - Radio communication
  - Scheduling
  - Handover

🔷 2. EPC (Evolved Packet Core)

EPC is the **core network** responsible for data and control.

🔸 (a) MME (Mobility Management Entity)

- Controls **signalling and mobility**
- Functions:
  - Authentication of users
  - Tracking user location
  - Session management

🔸 (b) S-GW (Serving Gateway)

- Acts as a **data router**
- Functions:
  - Forwards user data packets
  - Maintains data during handover
  - Connects eNodeB and P-GW

🔸 (c) P-GW (Packet Data Network Gateway)

- Connects LTE network to **external networks (Internet)**
- Functions:
  - Allocates IP address
  - Controls data usage and QoS
  - Acts as exit/entry point

🔸 (d) HSS (Home Subscriber Server)

- Central **database of users**
- Stores:
  - Subscriber details
  - Authentication information

- Service profiles

🔷 3. Interfaces in LTE

Interface	Connects	Function
S1-MME	eNodeB ↔ MME	Signalling
S1-U	eNodeB ↔ S-GW	Data transfer
S5/S8	S-GW ↔ P-GW	Data + control
S6a	MME ↔ HSS	Authentication
SGi	P-GW ↔ Internet	External connectivity

🔷 4. Traffic Types

- **User Traffic** → Actual data (internet, calls)
- **Signalling Traffic** → Control messages

🔷 5. Special Concepts (from diagram)

- **Local Breakout** → Data goes directly to internet (faster)
- **Home Routed Traffic** → Data routed through home network (secure)

🧠 Quick Summary (Exam Tip)

- **eNodeB** → Radio communication
- **MME** → Control & authentication
- **S-GW** → Data routing
- **P-GW** → Internet connection
- **HSS** → User database

📡 Multiplexing Techniques

🔹 Definition

Multiplexing is a technique used to **combine multiple signals into a single communication channel** to improve bandwidth utilization and reduce cost. The reverse process is called **demultiplexing**.

🔸 1. Frequency Division Multiplexing (FDM)

📌 Concept:

- The available bandwidth is divided into **multiple frequency bands**
- Each signal is transmitted **simultaneously on different frequencies**

⚙️ Features:

- Used for **analog signals**
- Requires **guard bands** to prevent overlap

✅ Advantages:

- Continuous transmission
- No delay between signals

❌ Disadvantages:

- Bandwidth wastage due to guard bands
- Susceptible to noise

📍 Applications:

- Radio broadcasting
- Cable TV

🔸 2. Time Division Multiplexing (TDM)

📌 Concept:

- Each signal is assigned a **specific time slot**
- Signals transmit **one after another in sequence**

⚙️ Types:

- **Synchronous TDM** → Fixed time slots
- **Asynchronous (Statistical) TDM** → Dynamic slots based on demand

✅ Advantages:

- Efficient bandwidth utilization
- No need for guard bands

❌ Disadvantages:

- Requires synchronization
- Delay may occur

📍 Applications:

- Telephone systems
- Digital communication

🔸 3. Wavelength Division Multiplexing (WDM)

📌 Concept:

- Used in **optical fiber communication**
- Multiple signals are transmitted using **different wavelengths (colors) of light**

⚙️ Types:

- **CWDM (Coarse WDM)**
- **DWDM (Dense WDM)**

✅ Advantages:

- Very high data transmission rates
- Efficient use of optical fiber

❌ Disadvantages:

- High cost
- Complex technology

📍 Applications:

- Fiber optic communication
- Internet backbone

🔸 4. Code Division Multiplexing (CDM / CDMA)

📌 Concept:

- All users share the **same frequency and time**
- Each signal is encoded with a **unique code**

⚙️ Features:

- Based on **spread spectrum technique**

✅ Advantages:

- High security
- Resistant to interference and noise

❌ Disadvantages:

- Complex system
- Requires strict synchronization

📍 Applications:

- Mobile communication (3G)
- GPS systems

🔷 Comparison Table

Technique	Basis	Key Idea	Signal Type
FDM	Frequency	Different frequencies	Analog
TDM	Time	Different time slots	Digital
WDM	Wavelength	Different light wavelengths	Optical
CDM	Code	Unique codes	Digital

🧠 Quick Summary (Exam Trick)

👉 **F → T → W → C = Frequency, Time, Wavelength, Code**

⚙️ Optimization

🔹 Definition

Optimization is the process of **improving the performance, efficiency, or quality** of a system by making the best use of available resources while satisfying given constraints.

🔸 Objectives of Optimization

- Maximize performance (speed, throughput)
- Minimize cost, delay, or energy consumption
- Efficient utilization of resources
- Improve Quality of Service (QoS)

🔸 Types of Optimization

1. Network Optimization

- Improves performance of communication networks
- Example:
    - Reducing congestion
    - Improving signal strength
    - Load balancing

---

2. Code Optimization

- Improves program efficiency
- Performed by compiler or programmer

📌 Types:

- **Machine-independent optimization** (loop optimization, dead code removal)
- **Machine-dependent optimization** (register allocation)

---

3. Database Optimization

- Improves query performance
- Example:
  - Indexing
  - Query rewriting
  - Normalization

---

4. Resource Optimization

- Efficient allocation of:
  - CPU
  - Memory
  - Bandwidth

🔸 Optimization Techniques

- **Heuristic methods** → Approximate solutions
- **Linear Programming** → Mathematical approach
- **Greedy algorithms** → Local optimum selection
- **Dynamic Programming** → Optimal substructure

🔸 Steps in Optimization

1. Define the problem
2. Identify objective function
3. Identify constraints
4. Apply optimization technique
5. Evaluate results

🔸 Advantages

- Improves system efficiency
- Reduces cost and time
- Enhances performance
- Better resource utilization

🔸 Disadvantages

- Complex process
- May require high computation
- Optimal solution not always guaranteed

🧠 Quick Summary (Exam Tip)

👉 **Optimization = Best solution under constraints**

📶 Wi-Fi Security Protocols

🔹 Introduction

Wi-Fi security protocols are used to **protect wireless networks** from unauthorized access, data theft, and attacks. They provide **authentication, encryption, and data integrity**.

🔸 1. WEP (Wired Equivalent Privacy)

📌 Overview:

- First Wi-Fi security protocol (IEEE 802.11)
- Uses **RC4 encryption algorithm**
- Key sizes: **64-bit / 128-bit**

⚙️ Features:

- Static key (same key for all users)
- Uses Initialization Vector (IV)

✅ Advantages:

- Simple to implement

❌ Disadvantages:

- Very weak security
- Easily hackable due to **IV reuse and weak encryption**

📍 Status:

❌ **Obsolete (not recommended)**

🔸 2. WPA (Wi-Fi Protected Access)

📌 Overview:

- Introduced as an improvement over WEP
- Uses **TKIP (Temporal Key Integrity Protocol)**

⚙️ Features:

- Dynamic key generation
- Message Integrity Check (MIC)

✅ Advantages:

- More secure than WEP
- Backward compatible

❌ Disadvantages:

- Still vulnerable to attacks
- Uses RC4 (outdated)

📍 Status:

⚠️ **Deprecated**

🔸 3. WPA2 (Wi-Fi Protected Access 2)

📌 Overview:

- Based on **IEEE 802.11i standard**
- Uses **AES encryption (Advanced Encryption Standard)**

⚙️ Features:

- Strong encryption
- Two modes:
    - **Personal (PSK)** → Password-based
    - **Enterprise** → Uses authentication server (RADIUS)

✅ Advantages:

- Highly secure
- Widely used

❌ Disadvantages:

- Vulnerable to:

- Weak passwords
- KRACK attack

📍 Status:

✅ **Still widely used**

🔸 4. WPA3 (Wi-Fi Protected Access 3)

📌 Overview:

- Latest Wi-Fi security protocol
- Introduced to overcome WPA2 weaknesses

⚙️ Features:

- Uses **SAE (Simultaneous Authentication of Equals)**
- Stronger encryption and protection

✅ Advantages:

- Better protection against password guessing
- Forward secrecy
- Strong encryption even with weak passwords

❌ Disadvantages:

- Requires modern hardware
- Not supported on older devices

📍 Status:

✅ **Recommended (latest standard)**

🔹 Comparison Table

Protocol	Encryption	Security Level	Status
WEP	RC4	Very Low	Obsolete

WPA	RC4 + TKIP	Medium	Deprecated
WPA2	AES	High	Widely Used
WPA3	SAE + AES	Very High	Latest

🔷 Key Concepts

🔐 Authentication

- Verifies user identity
- Example: Password, RADIUS server

🔐 Encryption

- Protects data from attackers
- Example: AES

🔐 Integrity

- Ensures data is not altered
